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# When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark

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**Abstract:**

LIME and SHAP are the most widely used model-agnostic explainers for tabular models, yet controlled paired comparisons of the two remain scarce. This paper extends an earlier paired test to the full grid of a public benchmark: 75 cells that share model, seed and sample size, over five model families on the UCI Adult census-income dataset, with Holm-corrected paired tests and effect sizes for five endpoints. SHAP leads on stability ($d_z$ = 3.00), fidelity ($d_z$ = 4.82) and faithfulness gap ($d_z$ = 2.63), and LIME gives sparser explanations. Two results depend on conditions that are usually left implicit. First, the latency ordering depends on the model family: SHAP's TreeExplainer was faster than LIME in every XGBoost cell and slower in every random-forest cell, and LIME was faster for non-tree models (median 53.3 ms vs. 694.6 ms). Second, LIME's near-zero stability on Adult is not a fixed property of the method: it is high on two further tabular datasets and, in a single-model probe, rises with the kernel width. The results support choosing the explainer according to the model architecture and the latency budget, and they show that a ranking obtained on one dataset and one configuration does not transfer by default. All artifacts are public.

***Keywords***: *Benchmarking, Explainable artificial intelligence, Explanation evaluation, LIME, SHAP*

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## 1. INTRODUCTION

LIME [1] and SHAP [2] are the two most widely used model-agnostic post-hoc explainers for tabular models. A practitioner who must choose one of them needs to know how they differ in the properties that matter for the deployment: whether an explanation tracks the model, whether it is stable, how many features it uses and how long it takes. Their theoretical differences are well known, but they do not by themselves settle the choice across model families, sampling regimes and runtime constraints [3, 4].

The empirical guidance is weaker than the popularity of the two methods suggests. Evaluation of explanations is spread over several metric families, and comparisons between explainers often depend as much on the metric chosen as on the methods themselves [5–7]. Work that uses several metrics shows that fidelity, stability, sparsity and cost carry complementary signals [8, 9]. Controlled comparisons of LIME and SHAP under matched conditions, where the two explainers see the same model, the same instances and the same random seed, remain scarce. Without that pairing, a difference between the methods cannot be separated from a difference between the models or the samples they were run on.

This paper reports such a paired comparison. It reuses the execution cohort released with an earlier study [10], which introduced the measurement protocol and compared four explainer families. That study already contains a first paired test of SHAP against LIME, on the 45 matched cells of three model families (logistic regression, random forest and XGBoost): SHAP was higher on fidelity, stability and faithfulness gap, and LIME was cheaper and sparser. It announced the analysis on the full grid of 75 cells without reporting it, and it reported averages over conditions only. The present paper completes that contrast on the 75 cells of the five model families, with confidence intervals and effect sizes, and then asks the question that an average over conditions hides: *when does the answer depend on the model and on the configuration?*

The paper makes three contributions:

1. **Paired inferential evidence on the full grid.** The SHAP–LIME difference, tested before on a 45-cell subset, is quantified on the 75 matched cells of the UCI Adult census-income dataset, over five model families, with Holm-corrected paired tests, confidence intervals and effect sizes for five endpoints (Section 4).
2. **Two conditional results.** The latency ordering of the two methods depends on the model family, and reverses between two tree ensembles (Section 4.4). LIME's near-zero stability on Adult depends on the kernel width and on the feature space, and is high on two further datasets (Sections 4.5 and 4.6).
3. **A deployment recommendation conditioned on the model architecture**, with an explicit statement of which claims generalize and which do not (Section 5).

The paper makes ranking claims under a stated protocol. It does not claim that either method is correct for a task, and it does not test whether users decide better with either explainer.

## 2. BACKGROUND AND RELATED WORK

### 2.1. LIME

LIME approximates a black-box model $f$ locally around an instance $x$ by training an interpretable surrogate $g$ on perturbed samples weighted by a kernel $\pi_x$, as in Eq. (1):

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{1}$$

where $\xi(x)$ is the explanation returned for $x$, $G$ is the class of interpretable surrogates (sparse linear models here), $\mathcal{L}$ measures how poorly $g$ reproduces $f$ on perturbed samples, each sample weighted by its proximity $\pi_x$ to $x$, and $\Omega(g)$ penalizes surrogate complexity [1]. LIME's behavior is shaped by the hyperparameters that build the neighborhood: the kernel width, the number of perturbed samples and the feature budget. These controls yield compact explanations and low latency, but they introduce stochastic variation, because local neighborhoods are sampled rather than exhaustively enumerated.

### 2.2. SHAP

SHAP explains a prediction through additive feature attributions $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$, where $\phi_0$ is the model's expected output over a background sample, $M$ is the number of features and $\phi_i$ is the Shapley-value contribution of feature $i$. SHAP values are the unique additive attributions that satisfy local accuracy, missingness and consistency [2]. KernelSHAP estimates them with a weighted regression over feature coalitions and applies to any model; tree-specific implementations exploit the model structure for speed. Relative to local-surrogate methods, SHAP is typically denser and, outside tree models, computationally heavier, because many coalition evaluations are required.

### 2.3. Evaluating and Comparing Explainers

Quantitative evaluation of explanations relies mostly on proxy metrics that need no human judgment: fidelity or faithfulness under feature removal [11–13], robustness of the explanation to small changes of the input [14], and structural properties such as sparsity. Toolkits and benchmarks standardize these measurements [15, 16], and systematic reviews document how unevenly they are applied [5, 6]. Several results warn against reading a proxy score as correctness: attribution methods can pass perturbation tests and fail sanity checks [17], removal-based scores depend on the removal scheme [18], and LIME and SHAP can both be manipulated [19]. Shapley-value attributions have known limitations as importance measures when features are dependent [20].

For a fair comparison of LIME and SHAP, confounders unrelated to the explainer must be controlled. This motivates the protocol of Section 3: an identical transformed feature space, shared model artifacts, a fixed class target, and cells matched on $(\mathrm{model}, \mathrm{seed}, N)$. Under these controls the expected trade-offs become testable: LIME should favor sparsity and speed, and SHAP should favor stability and faithfulness.

## 3. MATERIALS AND METHODS

### 3.1. Cohort and Research Questions

The runs analysed here come from the execution cohort released by Herrera-Vásquez and Herrero-Uceda [10], under the protocol and reproducibility contract defined there. The analysis is not the same: that study reports an omnibus comparison across four explainer families at the block level and a paired SHAP–LIME test restricted to 45 cells of three model families. The present analysis covers the 75 matched cells of the five families at the run level, with directional hypotheses, Holm-corrected paired tests, confidence intervals and per-endpoint effect sizes, and it examines how the differences vary with the model family, the dataset and the LIME configuration. Reusing the cohort is deliberate: a fresh execution would add environment variation to a comparison whose purpose is to isolate the explainer.

Four directional questions are addressed, each with its hypothesis, where $S$ is stability, $P$ sparsity, $C$ cost and $F$ fidelity:

- **RQ1 (robustness).** Is SHAP more stable than LIME under input perturbation? $H_1: S_{\mathrm{SHAP}} > S_{\mathrm{LIME}}$.

- **RQ2 (parsimony).** Is LIME sparser than SHAP? $H_2: P_{\mathrm{LIME}} < P_{\mathrm{SHAP}}$, where $P$ is the active-feature ratio (lower is sparser).
- **RQ3 (efficiency).** Is LIME computationally cheaper than SHAP? $H_3: C_{\mathrm{LIME}} < C_{\mathrm{SHAP}}$.
- **RQ4 (faithfulness).** Does SHAP give stronger fidelity and faithfulness signals than LIME? $H_4: F_{\mathrm{SHAP}} > F_{\mathrm{LIME}}$.

### 3.2. Experimental Design and Analysis Units

The benchmark has four factors: model family $g \in \{\mathrm{logreg}, \mathrm{rf}, \mathrm{xgb}, \mathrm{svm}, \mathrm{mlp}\}$; explainer $k \in \{\mathrm{LIME}, \mathrm{SHAP}\}$; seed $s \in \{42,123,456,789,999\}$; and sampling intensity $N \in \{50,100,200\}$ instances per stratum of the confusion matrix (TP, TN, FP, FN). This yields $5 \times 2 \times 5 \times 3 = 150$ planned runs, all of which are artifact-qualified (75 LIME, 75 SHAP), forming 75 matched SHAP–LIME cells on identical $(g, s, N)$ coordinates.

To avoid pseudoreplication, the analysis is hierarchical: metric values are computed per instance, averaged per run, and the paired SHAP–LIME run means on shared $(g, s, N)$ cells are the inferential unit.

### 3.3. Data, Models and Explainer Protocol

All experiments of the main cohort use the UCI Adult dataset [21, 22] with a stratified split and deterministic seeds. The preprocessing pipeline is fitted on the training partition only, with numeric scaling and one-hot encoding of categorical features, which gives 103 transformed features. A fixed set of pre-trained models is shared across all explainer runs to isolate the explainer effect: random forest [23], gradient-boosted trees [24], logistic regression, a support vector machine (SVM) and a multilayer perceptron (MLP). Evaluation instances are sampled by confusion-matrix stratum. Realized instance counts range from 27 to 800 (median 400).

Both explainers consume the same transformed feature space and target the probability of the positive class:

- **SHAP:** `TreeExplainer` for tree models and `KernelExplainer` for the other models, with a background sample of $n = 50$.
- **LIME:** tabular local surrogate with a feature budget of `num_features=10`, an effective `num_samples=1000` and `kernel_width=3.0`.

These are fixed reference settings, not the library defaults. The sensitivity of the results to the LIME kernel width is examined in Section 4.6.

### 3.4. Metrics

The metrics follow the protocol of the earlier study [10]. Throughout, $f(x)$ is the model's predicted probability of the positive class, $e(x) \in \mathbb{R}^M$ is the attribution vector an explainer returns for instance $x$ with $M$ features, and "masking" feature $j$ replaces its value with the training-set mean $\bar{x}_j$.

1. **Stability** $S$: robustness of the explanation to small input noise [14]. The explainer is run on $T = 15$ perturbed copies $x^{(a)} = x + \varepsilon^{(a)}$, with $\varepsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$ and $\sigma = 0.1$, and $S$ is the mean cosine similarity over all $T(T-1)/2$ pairs of the resulting attribution vectors $e^{(a)} = e(x^{(a)})$, Eq. (2):

$$S = \frac{2}{T(T-1)} \sum_{1 \le a < b \le T} \cos\left(e^{(a)}, e^{(b)}\right). \tag{2}$$

- $S$ is 1 when every perturbation yields an attribution vector with the same direction, and near 0 when directions are unrelated.

2. **Sparsity** $P$: active-feature ratio, the share of the $M$ features whose absolute attribution exceeds the threshold $10^{-4}$. A lower active ratio means a sparser explanation.
3. **Fidelity** $F$: a single-feature variant of the faithfulness correlation of Bhatt et al. [11]. Each feature $j$ is masked on its own, giving the masked instance $\tilde{x}^{(j)}$, and $F$ is given by Eq. (3):

$$F = \rho\left(\{|e_j(x)|\}_{j=1}^{M},\ \{|f(x) - f(\tilde{x}^{(j)})|\}_{j=1}^{M}\right), \quad (3)$$

- where $\rho$ is the Pearson correlation across the $M$ features. $F$ is high when the features an explanation ranks as important are those whose removal moves the prediction most.

4. **Faithfulness gap** $\Delta_k$: a deletion test in the spirit of [12, 13]. The $k = 5$ features with the largest $|e_j(x)|$ are masked together, giving $\tilde{x}^{(\text{top-}k)}$, and $\Delta_k$ is given by Eq. (4):

$$\Delta_k = \left|f(x) - f\left(\tilde{x}^{(\text{top-}k)}\right)\right|. \quad (4)$$

- A larger $\Delta_k$ means the features the explanation names first carry more of the prediction.

5. **Cost** $C$: per-instance wall-clock explanation time (ms).

LIME's active-feature count is bounded by `num_features=10`, which is a hard ceiling on the number of non-zero attributions. LIME's sparsity is therefore a consequence of a hyperparameter, not an emergent property of the method. The active-feature ratio measures whether LIME, *as configured*, returns fewer significant features than SHAP; it does not test whether LIME is intrinsically sparser.

### 3.5. Inferential Protocol

The statistical plan follows established practice for comparing learning methods [25]: a two-sided paired Wilcoxon signed-rank test on the matched cells for each metric [26]; the paired effect size $d_z$ for magnitude [27]; and Holm–Bonferroni control of multiplicity over the five primary metrics [28]. All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing.

### 3.6. Reproducibility and Computing Infrastructure

The protocol is configuration-driven (YAML files with seeded execution). Each run writes a `results.json` artifact; malformed or empty runs are excluded explicitly and nothing is imputed. All experiments ran on CPU. The reference software environment is Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, NumPy 2.2.6, pandas 2.3.3 and SciPy 1.16.3, pinned in the repository. Runs were executed on several commodity workstations under macOS, Windows and Linux, distributed by a work queue. Each run artifact records its configuration, seed, timestamps and duration; processor and memory specifications were not recorded. The consequence for the cost endpoint is discussed in Section 5.3.

## 4. RESULTS

The analysis covers 150 runs (75 LIME, 75 SHAP) forming 75 paired cells with identical $(\text{model}, \text{seed}, N)$. Matched coverage spans the five model families ($\text{xgb} = 15$, $\text{logreg} = 15$, $\text{rf} = 15$, $\text{mlp} = 15$, $\text{svm} = 15$) and the three sampling intensities.

### 4.1. Context: Earlier Results on This Cohort

The omnibus comparison over this cohort—SHAP, LIME, Anchors [29] and DiCE [30], tested over $n = 15$ blocks per method—is reported in the earlier study [10] and is not reproduced here. It establishes that the four methods are not exchangeable on fidelity or stability, that SHAP leads both endpoints, and that the SHAP–LIME fidelity separation falls marginally below the Nemenyi critical difference. Block aggregation was necessary there because Anchors and DiCE are not complete across the grid (57/75

and 68/75 artifact-qualified runs, respectively, against 75/75 for SHAP and LIME). The same study reports a paired Wilcoxon test of SHAP against LIME on 45 matched cells (logistic regression, random forest and XGBoost), significant on the five metrics in the directions found below. The analysis that follows extends that test to the 75 cells of the five model families. The 45 cells are a subset of the 75, so the two analyses are not independent. The earlier material is context, not evidence, for any claim made in this paper.

### 4.2. Primary Paired Inference

Table 1 summarizes the paired run-level outcomes on the 75 matched cells. The five primary endpoints are significant under Holm-adjusted testing.

*Table 1. Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, $\Delta$ = SHAP − LIME, with a 95% confidence interval on the mean paired difference. Sparsity is measured as active-feature ratio, so a positive $\Delta$ means SHAP produces denser explanations. Per-method mean levels for this cohort are reported in [10] and are not restated here*

| Metric | $\Delta$ | 95% CI | $p_{\text{Holm}}$ | $d_z$ |
|---|---|---|---|---|
| Stability | +0.718 | $[+0.663, +0.773]$ | $2.6 \times 10^{-13}$ | 3.00 |
| Sparsity (active ratio) | +0.142 | $[+0.103, +0.181]$ | $2.6 \times 10^{-13}$ | 0.84 |
| Fidelity | +0.248 | $[+0.236, +0.260]$ | $2.6 \times 10^{-13}$ | 4.82 |
| Faithfulness Gap | +0.045 | $[+0.041, +0.049]$ | $2.6 \times 10^{-13}$ | 2.63 |
| Cost (ms) | +8047.6 | $[+3939.2, +12155.9]$ | $1.2 \times 10^{-10}$ | 0.45 |

The stability effect is the largest separation in the study. The median paired stability difference is $+0.866$ in favor of SHAP, and the direction is the same in all 75 matched cells. Under the reference configuration, LIME's attribution vectors change direction under small input noise, whereas SHAP's are comparatively invariant.

For sparsity the sign is also the same in all 75 matched cells. The metric is the active-feature ratio, so SHAP's larger value means denser explanations: LIME is systematically sparser, within the limit stated in Section 3.4.

Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of $+0.248$ and a very large effect size ($d_z = 4.82$). The faithfulness gap favors SHAP in 74 of 75 matched cells, with a mean paired difference of $+0.045$ and a large effect size ($d_z = 2.63$). Figure 1 shows the paired mean differences for the four quality-oriented endpoints.

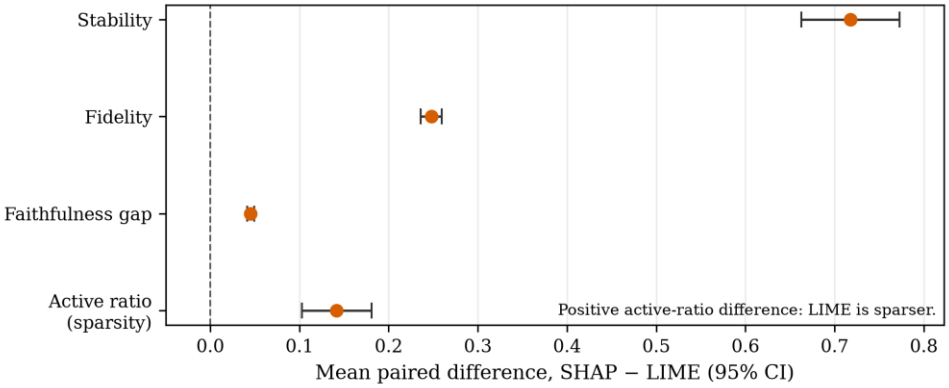



*Figure 1. Mean paired difference (SHAP − LIME) with its 95% confidence interval for the quality-oriented primary endpoints ($n = 75$ matched cells), as in Table 1. For the active ratio, a positive difference means SHAP is denser, that is, LIME is sparser*

### 4.3. Reproducibility across Seeds

To characterize how much the readings vary with the seed, the coefficient of variation (CV = SD/mean) was computed across the five seeds of the random-forest stratum at $N = 100$ (Table 2); the medians over all fifteen $(N, \text{model})$ strata are similar (0.9, 1.4, 2.3 and 88.2% in the table's row order). SHAP yields highly reproducible fidelity and stability readings ($\text{CV} < 1\%$). LIME fidelity is acceptable ($\text{CV} = 2.6\%$), whereas LIME stability has $\text{CV} = 86.2\%$. That figure mostly reflects the denominator: when the mean cosine similarity is near zero, any nonzero variation produces an extreme relative deviation. In absolute terms the spread is small (seed SD 0.015): the informative result is the level itself, which is near zero under the reference configuration.

*Table 2. Coefficient of variation (CV = SD/mean) across the five seeds of the random-forest, $N = 100$ stratum for the two primary quality endpoints. The CV is uninformative for LIME stability, whose mean is near zero*

| Method – Metric | CV | Reproducible? |
|---|---|---|
| SHAP – Fidelity | 0.8% | Yes |
| SHAP – Stability | 0.7% | Yes |
| LIME – Fidelity | 2.6% | Yes |
| LIME – Stability | 86.2% | CV not meaningful |

### 4.4. Runtime: the Ordering Depends on the Model Family

The cost difference is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. The matched-cell median cost is 684.1 ms for SHAP against 65.7 ms for LIME, and the median per-cell SHAP/LIME cost ratio is approximately 5.3 ×. The SHAP distribution is heavy-tailed, so its mean lies far above its median (the per-method means are reported in [10]): the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively, compared with 12,825 ms and 13,192 ms for LIME. This tail is concentrated in KernelSHAP contexts, especially the SVM.

The quality ordering (stability, sparsity direction, fidelity, faithfulness gap) is the same in every model family. The cost ordering is not. SHAP is faster in all 15 XGBoost matched cells and in one SVM cell, whereas the logistic regression, random forest and MLP cells are uniformly SHAP-slower (Figure 2, Table 3). The two tree ensembles behave in opposite ways. For XGBoost, `TreeExplainer` was faster than LIME in all 15 matched cells (median 18.1 ms vs. 61.8 ms). For random forests it was slower in all 15 cells (median 1,262.4 ms vs. 335.6 ms). A candidate explanation, not tested here, is that the cost of `TreeExplainer` grows with the number and depth of the trees. For the non-tree models, where SHAP runs in kernel mode, LIME has the lower median runtime (53.3 ms vs. 694.6 ms). "LIME is faster" is therefore a majority statement in this cohort, not a property of the two methods, and "SHAP is fast on tree models" does not hold for tree models in general.

*Table 3. Median per-instance cost by model group on the matched cells, and the number of cells in which SHAP was the faster method. SHAP uses `TreeExplainer` for the two tree ensembles and `KernelExplainer` for the non-tree models*

| Model group | Cells | SHAP median (ms) | LIME median (ms) | SHAP faster |
|---|---|---|---|---|
| XGBoost | 15 | 18.1 | 61.8 | 15 of 15 |
| Random forest | 15 | 1,262.4 | 335.6 | 0 of 15 |
| Non-tree (logreg, SVM, MLP) | 45 | 694.6 | 53.3 | 1 of 45 |

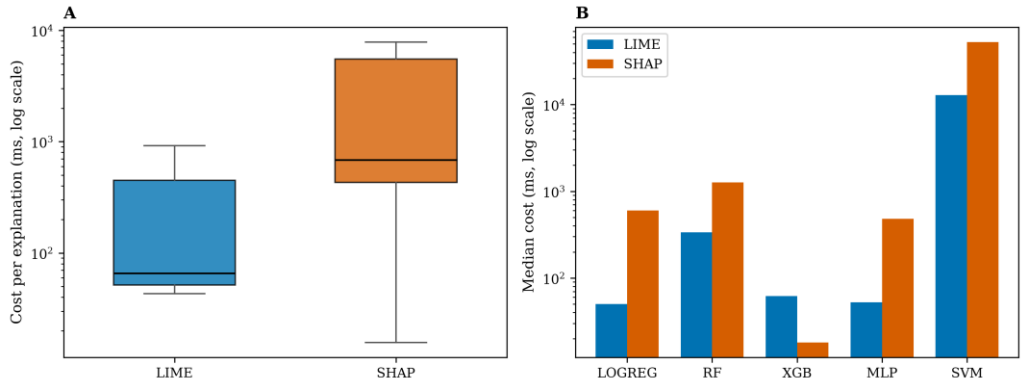



*Figure 2. Runtime behavior under matched-cell pairing ($n = 75$). (A) Overall cost distribution by method (log scale). (B) Model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower*

### 4.5. Cross-Dataset Extension

To probe external validity within tabular classification, a cross-dataset study (EXP3) compared SHAP, LIME and Anchors [29] on two further binary classification datasets: Breast Cancer Wisconsin [31, 32] ($n = 569$, 30 continuous features) and German Credit [31, 33] ($n = 1{,}000$, 20 mixed features). Both used random forest and XGBoost models across three random seeds each, yielding 12 paired dataset–model–seed comparisons per method pair.

**Fidelity ordering (SHAP vs. Anchors).** SHAP exceeds Anchors in fidelity in all 12 paired combinations (Table 4), with mean gaps ranging from $+0.26$ (German Credit, XGB) to $+0.51$ (Breast Cancer, RF).

*Table 4. Cross-dataset extension: mean Anchors fidelity (3-seed average) and the paired SHAP−Anchors fidelity gap. The gap is positive in all 12 paired combinations. SHAP fidelity levels for this cohort are reported in [10]; the Anchors levels and the gaps are contributed here*

| Dataset | Model | Anchors | Gap (SHAP − Anchors) |
|---|---|---|---|
| Breast Cancer | RF | 0.265 | $+0.514$ |
| Breast Cancer | XGB | 0.208 | $+0.409$ |
| German Credit | RF | 0.351 | $+0.363$ |
| German Credit | XGB | 0.451 | $+0.260$ |

**LIME on the further datasets.** Table 5 reports LIME fidelity and stability on the same datasets, with the paired SHAP−LIME fidelity gap. Two findings follow. First, SHAP > LIME fidelity holds on German Credit (mean gaps $+0.23$ for RF and $+0.24$ for XGB, positive for every seed), in the direction found on Adult. On Breast Cancer the fidelity gap narrows substantially: SHAP leads LIME by only $+0.07$ (RF) and $+0.07$ (XGB), compared to $+0.27$ (RF) and $+0.21$ (XGB) in the corresponding model strata on Adult. Second, LIME *stability* on these datasets (0.75–0.93) is an order of magnitude higher than on Adult, where it is near zero, at the same reference kernel width. The near-zero stability observed on Adult is therefore not a fixed property of LIME.

*Table 5. Cross-dataset extension: mean fidelity and stability (3-seed average) for LIME, and the paired SHAP−LIME fidelity gap. Stability is the mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$), the protocol*

*of the main cohort. For comparison, LIME on Adult in the matched model strata: fidelity* $0.457$ *(RF) and* $0.553$ *(XGB); stability* $\approx 0.01$–$0.02$ *across all model families*

| Dataset | Model | Fidelity | Gap (SHAP − LIME) | Stability | Sparsity | Cost (ms) |
|---|---|---|---|---|---|---|
| Breast Cancer | RF | 0.713 | $+0.07$ | 0.922 | 0.331 | 12.3 |
| Breast Cancer | XGB | 0.543 | $+0.07$ | 0.927 | 0.331 | 7.5 |
| German Credit | RF | 0.483 | $+0.23$ | 0.854 | 0.163 | 21.1 |
| German Credit | XGB | 0.468 | $+0.24$ | 0.748 | 0.163 | 10.2 |

SHAP stability was not instrumented in the cross-dataset study, so a direct SHAP–LIME stability comparison on Breast Cancer and German Credit is not available.

### 4.6. LIME Stability Depends on the Kernel Width

A probe on Adult varied the LIME kernel width with everything else fixed (random forest, seed 42, $N = 100$, `num_samples=1000`, `num_features=10`). Table 6 reports the result. Fidelity and stability are computed over the instances that have at least one non-zero weight. The probe covers one model and one seed, so it shows that the dependence exists; it does not measure its size across models.

*Table 6. LIME sensitivity to* `kernel_width` *on Adult (random forest, seed 42,* $N = 100$*). Reference condition:* `kernel_width=3.0`

| `kernel_width` | Valid instances | Fidelity | Stability | Sparsity | Cost (ms) |
|---|---|---|---|---|---|
| 0.25 | 0/100 (degenerate) | — | — | 0.000 | 30 |
| 0.75 | 0/100 (degenerate) | — | — | 0.000 | 30 |
| 3.0 *(ref.)* | 96/100 | 0.518 | 0.000 | 0.087 | 30 |
| 10.0 | 100/100 | 0.441 | 0.664 | 0.093 | 32 |

Kernel widths of 0.25 and 0.75 produce zero-weight explanations for all 100 instances, because the exponential neighborhood collapses in the 103-dimensional one-hot-encoded space. The reference width of 3.0 is the smallest of the tested settings that yields usable explanations. Widening the kernel to $10.0$ raises stability to $0.664$ at a cost in fidelity ($0.518 \rightarrow 0.441$). Together with the cross-dataset result, where LIME reaches $0.75$–$0.93$ at the reference width, the probe supports a bounded conclusion: under the reference kernel width, LIME's stability on Adult is near zero; it rises with the kernel width and is high on two lower-dimensional datasets, so it is modulated by the configuration and by the feature space. Adult's one-hot encoding is a candidate cause, and the stability protocol itself may contribute: the Gaussian noise is added to every transformed feature, including the one-hot indicator columns, which then take values that no real instance has. Breast Cancer has no categorical feature. No experiment here separates the encoding from the perturbation scheme, and the two further datasets also differ from Adult in size, class balance and the mix of continuous and categorical features, so this design does not isolate either cause. The near-zero reading should not be taken as a configuration-independent property of LIME.

### 4.7. Sensitivity of the Fidelity Ordering to the Masking Scheme

The fidelity and faithfulness-gap metrics mask features independently. Adult contains associated features, so a masked instance may fall outside the training distribution, where model outputs are less reliable. To assess whether this changes the method ordering, a probe reused the stored top-10 explanations of the five Adult random-forest runs at $N = 50$, one per seed, and recomputed the prediction-drop metrics under four masking schemes: independent zero masking, mean replacement, marginal empirical replacement, and grouped mean replacement, which masks associated feature sets together (for example `education` and `education-num`). Table 7 reports the result.

*Table 7. SHAP–LIME top-k prediction-drop advantage and drop-correlation advantage under four masking schemes (Adult random-forest stratum, five runs at $N = 50$, one per seed). The SHAP direction is positive in all five paired seed cells under every scheme*

| Masking scheme | Mean SHAP–LIME gap | Mean drop-correlation delta |
|---|---|---|
| Independent zero masking | +0.098 | +0.515 |
| Mean replacement | +0.098 | +0.512 |
| Marginal empirical replacement | +0.081 | +0.625 |
| Grouped mean replacement | +0.053 | +0.494 |

SHAP's advantage is positive in all five paired cells under every scheme. Its magnitude, measured as the prediction-drop gap, does not increase, and falls overall, as the masking takes more of the feature dependence into account ($+0.098 \rightarrow +0.098 \rightarrow +0.081 \rightarrow +0.053$). The drop-correlation delta does not follow this ordering: it is highest under marginal empirical replacement ($+0.625$) and lowest under grouped mean replacement ($+0.494$). With five pairs, the exact Wilcoxon test cannot reach $\alpha = 0.05$ under any scheme, so this agreement is descriptive. The probe uses stored top-10 explanations, not full attribution vectors, and is a stress test of the construct, not a replacement for the primary analysis: the direction holds in every pair, and the magnitude depends in part on the masking protocol.

## 5. DISCUSSION

### 5.1. What the Paired Analysis Adds

The four hypotheses are supported on Adult: SHAP is more stable (RQ1) and scores higher on fidelity and faithfulness gap (RQ4); LIME is sparser as configured (RQ2) and cheaper in most cells (RQ3). The analysis on the full grid confirms the direction that the earlier 45-cell test found, extends it to the SVM and the MLP, and attaches an interval and an effect size to each endpoint; the fidelity and stability differences have the same sign in every cell. SHAP delivers higher stability and fidelity at a substantially higher expected runtime; LIME delivers lower latency and more compact explanations at the cost of robustness.

Two of these results come with conditions, and the conditions are the part most likely to matter in practice. The runtime result (RQ3) reverses between XGBoost and random forests, two models that a guideline would group together as "tree-based". The stability result (RQ1) is measured at one LIME kernel width on one feature space; a wider kernel or a lower-dimensional dataset raises LIME's stability from near zero to between $0.664$ and $0.927$. A comparison run on one dataset with one configuration would report either of these as a property of the methods.

One candidate reason for SHAP's fidelity advantage is that SHAP ties attribution mass to the model's response to feature removal: local accuracy means the attributions sum to the prediction change relative

to the background distribution [2], and removal-based fidelity asks a related question. This mechanism was not tested and carries no guarantee: the masking used here replaces features with reference values and not with SHAP's background expectation, KernelSHAP is itself a sampled estimate, and the advantage is small on Breast Cancer.

### 5.2. Recommendation Conditioned on the Model Architecture

Table 8 states the recommendation in two tiers. The first-tier criterion is not a preference for a method. It is a function of the model architecture: where a model-aware explainer is also fast, as `TreeExplainer` was for XGBoost, the trade-off between fidelity and latency disappears and SHAP is the better choice in both tiers; where it is not, as for the random forests here, the trade-off remains and has to be measured for the deployed model.

*Table 8. Recommendation by deployment tier, conditioned on the model architecture*

| Tier | Recommendation | Evidence and condition |
|---|---|---|
| Tier 1: real-time, end-user facing | **SHAP** (`TreeExplainer`) for XGBoost; **LIME** for models without a fast model-aware explainer; for other tree ensembles, measure latency before choosing. | For XGBoost, SHAP was faster in all 15 matched cells and more faithful. For random forests it was slower in all 15 cells. For non-tree models LIME has the lower median runtime (Section 4.4). Applies when sub-second latency and compact explanations matter more than maximum fidelity. |
| Tier 2: auditing, model analysis | **SHAP.** | Consistent paired advantages in stability, fidelity and faithfulness gap ($d_z = 3.00, 4.82, 2.63$). Applies to offline or analyst-mediated work that tolerates latencies of seconds to minutes; the sample budget depends on whether SHAP runs in tree-specific or kernel mode. |

### 5.3. Validity and Limitations

**Ranking claims, not validity claims.** The differences reported here are comparative performance within a benchmark protocol. A claim that an explanation is correct for a task would need stronger evidence: ground truth from synthetic or transparent models, perturbation protocols that respect feature dependence, or human-centered validation [34, 35]. Formal analyses show that several attribution families assign high importance to suppressor variables when features are dependent [36, 37], and faithfulness-style evaluations are informative but do not settle correctness on their own [18, 38].

**Configuration.** Both methods were run with one fixed reference configuration. The cost gap reflects the method family and the concrete implementation pathway (tree-specific or kernel SHAP). The kernel-width probe covers one model and one seed. Sensitivity to the SHAP background sample and to the kernel width in other models was not examined.

**Execution hosts and the cost endpoint.** Cost is per-instance wall-clock time, and the runs were executed on several workstations whose processor and memory specifications were not recorded (Section 3.6). The two members of a paired cell are matched on model, seed and $N$, but did not necessarily run on the same host. Absolute latencies, and the medians reported here, include between-host variance and are indicative of this environment; they are not hardware-normalized measurements. The quality endpoints do not depend on the host.

**Generalization.** Table 9 states the boundary of each claim. The comparison is limited to tabular binary classification; image, text, graph and time-series explanations need different masking schemes and endpoints and were not tested. The endpoints measure how well an explanation tracks the model it

explains. They say nothing about whether that model is accurate or fair, and nothing about whether a person decides better with the explanation.

*Table 9. Generalizability boundary of the empirical claims*

| Claim | Evidence | Generalization status |
|---|---|---|
| SHAP > LIME stability | Adult matched cells, one LIME kernel width | Holds for the Adult protocol only; LIME stability is high on two further datasets, where SHAP stability was not measured |
| SHAP > LIME fidelity | Adult, plus German Credit and Breast Cancer | Supported within tabular classification, with a small gap on Breast Cancer; not established for other data types |
| LIME lower runtime | Adult, model-dependent | Bounded by the model architecture and the SHAP implementation; `TreeExplainer` reverses the ordering for XGBoost but not for random forests |
| SHAP > Anchors fidelity | Breast Cancer and German Credit | Fidelity ordering only; not a stability or runtime result |
| Usefulness for people | No participant study | Not established |

## 6. CONCLUSIONS

On 75 matched cells of the Adult dataset, SHAP shows consistent paired advantages over LIME in stability ($d_z = 3.00$), fidelity ($d_z = 4.82$) and faithfulness gap ($d_z = 2.63$), and LIME produces sparser explanations and runs faster in most model contexts. A cross-dataset extension supports the fidelity direction on two further tabular datasets.

The choice between the two methods is a matter of fit for purpose, and two of the usual rules of thumb need a condition. SHAP is not fast on tree models in general: it was faster than LIME for XGBoost and slower for random forests. LIME is not unstable in general: its near-zero stability on Adult is specific to the reference kernel width and to that feature space. The practical recommendation is to prefer SHAP where stability and fidelity dominate, to prefer LIME where latency dominates and no fast model-aware explainer exists, and to measure latency for the deployed model before relying on either rule.

Future work should instrument SHAP stability on further datasets, vary the SHAP background sample and the LIME kernel width across models, and extend the comparison beyond tabular classification. Whether either method helps people decide is a separate question that needs a study with participants.

### Competing Interests

No competing financial interests or personal relationships that could have appeared to influence the work reported in this paper are declared. No external funding was received for this study.

### Data and Code Availability

The experimental code, the run artifacts and the analysis exports are available in a public repository and in a versioned archive with a DOI, both released under the MIT License; their addresses are withheld for masked review and are given on the title page. The artifacts needed to reproduce this study are the main-cohort results (), the inferential exports (), the cross-dataset results ( and ), the kernel-width probe () and the masking-sensitivity exports (). Every reported value is re-derived from these files by a verification script that runs in continuous integration.

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